



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VI

Subject Name: Medicinal Chemistry-III

Subject Code: BP601T

Max. Marks: 75

Date: 13/07/2022

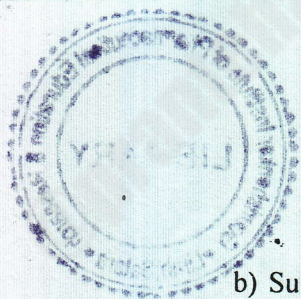
Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- i) Basic Nucleus of Cephalosporins and Penicillins is.....
 - a) Lactone
 - b) Thiazole
 - c) Lactam Ring
 - d) Beta Lactam Ring
- ii) Streptomycin is produced by which of the following organisms?
 - a) Streptomyces griseus
 - b) Streptomyces fradiae
 - c) Streptomyces noursei
 - d) Streptomyces nodosus
- iii) What is Target for clavulanic acid?
 - a) Lactamase
 - b) L-alaracemase
 - c) The transpeptidase enzyme
 - d) Penicillin acylase
- iv) Aminoglycosides work by irreversibly binding to the..... ribosomal subunit.
 - a) 50S
 - b) 30S
 - c) both
 - d) none
- v) Benzyl Penicillin is chemical name for which of the following penicillin..?
 - a) Penicillin-G
 - b) Penicillin-V
 - c) Penicillin-F
 - d) Phenethicin
- vi) Chemically tetracycline is derivative of.....
 - a) Purine
 - b) Pyridine
 - c) Octahydronaphthacene
 - d) Phenanthrene
- vii) Which is naturally occurring anti-malaria drug?
 - a) Quinacrine
 - b) Artemisinin
 - c) Mepacrine
 - d) Mefloquine
- viii) Chloroquine can be synthesized by reaction of 4-diethylamino-1-methylbutylamine with?
 - a) 4,7-dichloroquinoline
 - b) α -naphthol
 - c) β -naphthol
 - d) None of the above
- ix) What is co-trimoxazole?
 - a) Sulphamethoxazole + trimethoprim, 3:1



- b) Sulfadoxine + pyrimethamine
c) Sulphamethoxazole + trimethoprim, 5:1
d) Sulfacetamide, Sulfathiazine and sulfabenzamide
- x) Identification of a new chemical entity as a potential therapeutic agent (From Hit to Lead) is known as
a) Drug discovery
b) Drug development
c) Both of above
d) None of above
- xi) In QSAR, study of medicinal chemistry Q stands for
a) Qualitative
b) Quantitative
c) Both
d) Quantam
- xii) Identify the metabolite of prontosil responsible for its antibacterial activity?
a) Sulphacetamide
b) Sulphanilamide
c) PABA
d) Probenec
- xiii) A combination of medications which applied to treat tuberculosis is
a) 1-thia-4-azabicyclo[3.2.1]heptane
b) 4-thia-1-azabicyclo[3.2]heptane
c) 4-thia-1-azabicyclo[3.2.0]heptane
d) 1-thia-4-azabicyclo[1.2.3]heptane
- xiv) Which of the following is noncompetitive inhibitor of the enzyme reverse transcriptase in HIV?
a) Lamivudine
b) Nevirapine
c) Ritonavir
d) Tenofovir
- xv) Sulphonamides do not have adverse drug interaction with
a) Oral coagulants
b) Sulphonylureas
c) Hydantoin anticoagulants
d) Dihydrofolatereductase inhibitors
- xvi) The essential structural unit for the anthelmintic activity of mebendazole is:
a) Benzoyl group
b) Benzimidazole
c) Methyl carbamate
d) Imidazole
- xvii) Artemisinin contains the following group in its structure?
a) An endoperoxide
b) An exoperoxide
c) An epoxide
d) An acid hydrazide
- xviii) Identify the antibiotic obtained from streptomyces sp. but not having antibacterial property. In fact it is suicide inhibitor of beta lactamases.
a) Azetronam
b) Cilastatin
c) Clavulanic acid
d) Imipenem
- xix) Methotrexate is thought to exert its action by.....
a) Interfering with purine synthesis
b) Intracellular formation of an amine adduct
c) Forming a conjugate with nucleic acids
d) Inhibiting the synthesis of folic acid by inhibiting dihydrofolate synthetase
- xx) Chloramphenicol has two asymmetric centers. What is the configuration and optical activity of the active form of the antibiotics?
a) D-threo, (-)
b) L-threo, (+)
c) D- erythro, (-)
d) L- erythro, (+)

Q. 2. Long Answers

2 x 10 = 20 (Answer 2 out of 3)

1. What are antibiotics? Give Chemical Classification of antibiotics with structures. Discuss in brief about Chemistry and SAR of antibiotics containing Penam ring system.
2. Classify antimalarial agents. Discuss SAR and MOA of 4-aminoquinolines and 8-aminoquinolines. Outline the synthetic scheme for Chloroquine.
3. Classify Sulphonamides with structures. Write Chemistry and MOA of Sulpha drugs. Add note of DHFR inhibitor. Give the synthesis of Trimethoprim.

Q. 3. Short Answers

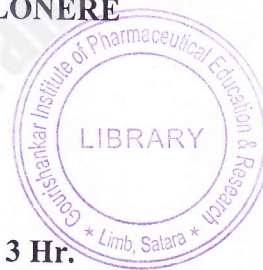
7 x 5 = 35 (Answer 7 out of 9)

- i) Discuss chemistry, SAR and MOA of Tetracycline.
- ii) What are anti-mycobacterial agents? Draw the structures of any two anti-tubercular drugs. Outline the synthesis of p-amino salicylic acid.
- iii) Write in brief about anthelmintics. Outline synthetic scheme for Mebendazole.
- iv) What are aminoglycosides? Write MOA and Chemistry of aminoglycosides.
- v) Define Prodrug. Give the Applications of Prodrug with examples.
- vi) Classify antiviral agents. Discuss in brief about RTI's.
- vii) Discuss chemistry, SAR, MOA of fluoroquinolone antibacterial agents giving synthetic of Ciprofloxacin.
- viii) Write a note on combinational chemistry & its applications in drug discovery.
- ix) Give chemical classification of antifungal drugs with example. Discuss chemistry of Imidazole antifungal

-----END OF THE PAPER-----

Course : B. Pharmacy
Subject Name : Medicinal Chemistry-III

Sem: VI
Subject Code :
BP601T
Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1 Objective type Questions (Answer all the questions)

10 x 2=20M

1. Define the term antibiotic with suitable example.
2. Write a note on- Macrolide antibiotic
3. Classify anti-tubercular agents.
4. Write about quinolones.
5. Comment on- Antifungal antibiotics.
6. Write about folate reductase inhibitors.
7. What are various approaches used in drug design.
8. Enlist different applications of combinatorial chemistry.
9. What are anthelmintics.
10. Define the term prodrugs.
11. Explain classification and SAR of Sulfonamides.
12. Explain etiology of malaria

Q2. Long Answer Questions. (2 Out of 3)

2 x 10=20M

1. Explain in brief about stereochemistry, structure activity relationship and uses of β -lactam Antibiotics.
2. Write a note on- Synthetic anti tubercular agents.
3. Explain in brief about Antiviral agents.

Q. 3 Short Answer Questions. (7 Out of 9)

7 x 5=35M

1. Explain Pharmacophore modeling and docking techniques.
2. Write a note on solid phase synthesis.
3. Explain in brief about dapsone.
4. Comment on- Biguanides and dihydro triazines.
5. Explain in brief about stereochemistry and structure activity relationship of Aminoglycosides.
6. Explain unclassified antibiotics with suitable examples.
7. Explain in brief about structure activity relationship of quinolones.
8. What are Tetracyclines?
9. Write in brief about Anti-protozoal Agents.

Course: B. Pharmacy

Sem: 6th

Subject Name: Medicinal Chemistry-III

Subject Code: BP601T

Max Marks: 75

Date: 10/07/2023

Duration: 3 Hr



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type questions (Answer all the questions)

(10 * 2) = 20

- i. Write the structure and uses of Penicillin-G.
- ii. Give the synthesis of Isoniazid.
- iii. List out important anti-viral agents. Draw the structure of acyclovir.
- iv. Write the structure of any two anti-malarial drugs.
- v. What are antiprotozoal agents? Write the structure of metronidazole.
- vi. What are anthelmintics? Give examples.
- vii. Give the structure and IUPAC name of Dapsone.
- viii. Write the synthesis of para amino salicylic acid.
- ix. Define anti-tubercular agents? Name the causative organism for tuberculosis.
- x. Enlist the parameters of QSAR.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i. Define and Classify antimalarial agents with examples. Give the mechanism of action and outline the synthesis of chloroquine.
- ii. What are anthelmintics? Classify with suitable examples. Outline the synthesis and mechanism of action of Mebendazole.
- iii. What are antibiotics? Classify with examples. Discuss the SAR & MOA of tetracyclines.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i. What are aminoglycosides? Write the mechanism and SAR of aminoglycoside antibiotics.
- ii. Write a note on urinary tract anti-infective agents.
- iii. What are Sulphonamides? Explain their SAR.
- iv. Write a note on combinatorial chemistry and its applications.
- v. Define and classify anti-tubercular agents with examples. Write the MOA of isoniazid.
- vi. Add a note on synthetic antifungal agents. Give the synthesis of Tolnaftate.
- vii. Write a note on prodrug concept and applications of prodrugs design.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Date : 30/12/2023
Course : B. Pharmacy
Subject Name : Medicinal Chemistry III
Max Marks : 75
Sem: VI
Subject Code: BP601T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Draw the structure of two tetracyclines.
- ii) Outline synthesis of chloroquine.
- iii) Enlist the physicochemical parameters used in QSAR.
- iv) What is the mechanism of action of isoniazid?
- v) What are the antimalarial derived from natural source?
- vi) Draw any two structures from quinolone.
- vii) Write examples the antifungal antibiotics.
- viii) Write down mechanism of action of metronidazole.
- ix) What is the category of Thiabendazole, Amodiaquine?
- x) Enlist applications of combinatorial chemistry.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Write in detail about classification of antibiotics depending on its chemical structure. Give emphasis on β -lactum antibiotics mechanism of action, examples.
- ii) Give detailed account on sulfonamides and sulfones with its nomenclature, mechanism of action, spectrum.
- iii) What is drug design? What are the various approaches used in drug designing detail?

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Write a note on combinatorial chemistry.
- ii) Discuss about benzimidazole derivatives as the antiprotozoal?
- iii) Give emphasis on reverse transcriptase inhibitor.
- iv) Write a note on agents used as urinary tract anti-infective.
- v) Enlighten aminoglycoside antibiotics.
- vi) Draw the synthesis of: Chloramphenicol and Para Amino Salicylic Acid.
- vii) Explain the basic concept of prodrugs and its applications.
- viii) Write a note on chemical classification of antimalarial agents.
- ix) What are synthetic antitubercular agents?

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: Medicinal Chemistry-III

Max Marks: 75

Date: 12-06-2024

Semester: VI

Subject Code: BP601T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Write down the applications of Prodrugs.
- ii) Give the structure and use of Clindamycin.
- iii) Write down about beta lactamase inhibitors.
- iv) Outline the synthesis of Para Amino Salicylic Acid.
- v) Discuss about folate reductase inhibitors.
- vi) Draw structure of Ciprofloxacin and Nitrofurantoin.
- vii) Discuss the MOA of aminoglycoside.
- viii) Draw any two structures from Imidazole derivatives.
- ix) Enlist the applications of Combinatorial Chemistry.
- x) Discuss the term partition coefficient.

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Give in detail chemical classification of antibiotics with example and structure. Discuss the SAR, MOA and uses of Cephalosporin derivatives.
- ii) Classify antimalarial agents with example and structure. Give SAR of 4-aminoquinolines. Outline the synthesis of Chloroquine.
- iii) Give classification of antiviral agents with example. Discuss MOA of purine analogues. Draw the synthesis of Acyclovir.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Write note on QSAR.
- ii) Discuss the antiprotozoal agents.
- iii) Give brief account of Tetracycline.
- iv) Outline synthesis of Chloramphenicol. Give its uses.
- v) Discuss agents used as urinary tract anti-infective.
- vi) Write down short note on macrolide antibiotic.
- vii) Enlighten beta lactam antibiotics.
- viii) Write note on Synthetic antitubercular agents.
- ix) Give an account of MOA and SAR of Sulphonamide.

*** END OF THE PAPER ***

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